

Students Name:	Grade: 6	Date:
Subject: Mathematics	Unit Title: Numerical Reasoning	
Task: Term 1 Practice sheet		Task Duration:

A Natural numbers, whole numbers and integers

1. Sort these numbers into the table below: -7 , 0 , $\frac{3}{4}$, 15 , -2.5 , 100 , 1 , -48

Natural numbers	Whole numbers	Integers	None of these

2. Write down the smallest natural number and the smallest whole number. _____

3. Draw a number line from -6 to 6 and mark: -4 , -1 , 0 , 3 , 5 .

4. Arrange in ascending order: -12 , 7 , 0 , -3 , 9 , -20 _____

5. Fill in $<$, $>$ or $=$:

a -8 ___ -3 b 0 ___ -5 c -15 ___ 15 d -1 ___ -100

B Factors, multiples and prime numbers

1. List all the factors of 36 . _____

2. Write the first five multiples of 8 . _____

3. Find all the common factors of 24 and 36 . _____

4. Write the first three common multiples of 4 and 6 . _____

5. Circle the prime numbers: 1 , 2 , 9 , 13 , 21 , 29 , 33 , 37 , 51 , 97

c Prime factorization, HCF and LCM

Use a factor tree or the division method. Write answers in index form where you can.

1. Write as a product of prime factors:
a 84 **b** 180 **c** 225 **d** 360
2. Find the HCF and the LCM of 24 and 36.
3. Find the HCF and the LCM of 45 and 60.
4. Find the HCF and the LCM of 18 and 48.

D Fractions and decimals

1. Simplify to lowest terms:

a $\frac{18}{24}$ b $\frac{45}{60}$ c $\frac{84}{144}$ d $\frac{96}{120}$

2. Change to decimals:

a $\frac{3}{8}$ b $\frac{7}{20}$ c $\frac{5}{16}$

3. Change to fractions in lowest terms:

a 0.45 b 0.125 c 2.6

4. Fill in <, > or =:

a $\frac{3}{5}$ — $\frac{5}{8}$ b $\frac{7}{12}$ — 0.58 c $\frac{4}{9}$ — 0.44

5. Arrange in ascending order: $\frac{2}{3}$, 0.7, $\frac{5}{8}$, 0.58

6. Find the missing number:

a $\frac{3}{7} = \frac{?}{49}$ b $\frac{5}{8} = \frac{45}{?}$

E**Adding and subtracting fractions and decimals**

1. $\frac{3}{8} + \frac{5}{12}$

2. $\frac{7}{10} - \frac{2}{15}$

3. $2\frac{3}{4} + 1\frac{5}{6}$

4. $5\frac{1}{3} - 2\frac{3}{4}$

5. $\frac{1}{2} + \frac{2}{3} + \frac{3}{4}$

6. $12.75 + 3.6 + 0.084$

7. $25 - 8.47$

8. $9.3 - 4.876$

F Order of operations (BODMAS)

Write one line of working for each step. Do not jump straight to the answer.

1. $36 \div 6 + 4 \times 3 - 5$

2. $48 \div (8 + 4) \times 3$

3. $15 + [24 \div (3 + 5)] \times 2$

4. $7 \times (12 - 5) + 18 \div 3^2$

5. $100 - \{40 + (18 \div 3 \times 2)\}$

6. $\frac{2}{3} \text{ of } 45 + 0.25 \times 40$